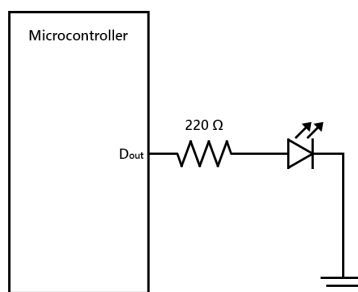


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|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Course</b>         | : Diploma in Electronic Systems (EGDF18)<br>Diploma in Electronic & Computer Engineering (EGDF20)                                        |
| <b>Module</b>         | : Embedded System Design & Technology (EGE320)                                                                                           |
| <b>Tutorial No</b>    | : Tutorial 6                                                                                                                             |
| <b>Tutorial Title</b> | : Digital Output and Input – Digital Output                                                                                              |
| <b>Objective</b>      | : Describe the use of digital output and input by identifying different types of external circuit connected to digital output and input. |

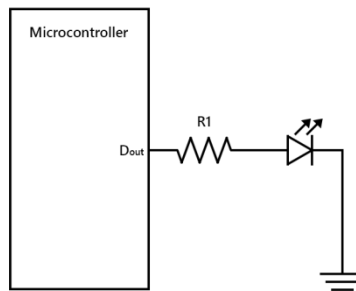
**Questions:**

- Which of the following voltage can be generated by a digital output in the microcontroller assuming the  $V_{DD}$  of the microcontroller is 3.3 V?
  - 3.29 V
  - 2.80 V
  - 1.65 V
  - 0.80 V
- Which of the following electronic devices is **NOT** suitable to be controlled by using a digital output.
  - LED light bulb.
  - Portable Fan.
  - Smart Door Lock.
  - Temperature Sensor.
- Calculate the current drawn by the LED given  $D_{out} = 3.30$  V and  $V_{LED} = 1.20$  V. Refer to the circuit below.
  - 5.46 mA
  - 9.55 mA
  - 15 mA
  - 20.46 mA

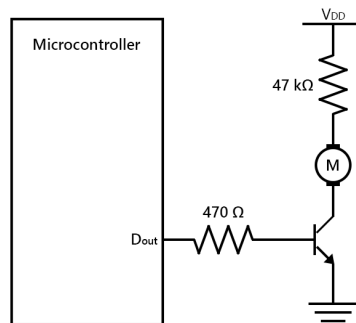


4. Calculate the resistance of R1 to limit the current drawn by the LED to 3 mA given  $D_{out} = 3.30\text{ V}$  and  $V_{LED} = 0.8\text{ V}$ . Refer to the circuit below.

- A.  $266.67\ \Omega$
- B.  $833.33\ \Omega$
- C.  $1.10\text{ k}\Omega$
- D.  $1.37\text{ k}\Omega$



5. Which of the following is the **PURPOSE** of the use of transistor in the circuit below?



- A. To isolate the external circuit from the microcontroller.
  - B. To protect microcontroller from being short-circuited.
  - C. To allow higher current flow to drive the motor.
  - D. To increase the component counts in the entire circuit.
6. Which of the following is one of the digital output type?
- A. Open-collector / drain
  - B. Pull-up
  - C. Pull-down
  - D. Floating
7. Which of the following digital output type are able to sink and source current?
- A. Open-collector / drain
  - B. Pull-up
  - C. Pull-down
  - D. Push-pull

8. What will happen to the digital output when the internal transistor of an Open-collector / drain output is turned off?
  - A. It sinks current to the internal ground through an internal transistor.
  - B. It sources current from the internal power supply through an internal transistor.
  - C. It is open with high impedance.
  - D. It is closed and shorted to ground.
9. What is the advantage of open-drain output over push-pull output?
  - A. Allows multiple devices to share the same line without damage. It relies on an external pull-up resistor to bring the line to the desired voltage.
  - B. Fast switching as the output is driven both way by transistors.
  - C. No pull-up resistor is required.
  - D. Several outputs from different devices cannot be connected directly together.
10. What is the advantage of push-pull output over open-drain output?
  - A. It can be pulled-up to any voltage found in the system.
  - B. Fast switching as the output is driven both way by transistors.
  - C. Pull-up resistor is required.
  - D. Several outputs from different devices can be connected directly together.